

Aniket Gupta

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EDUCATION

Stanford University — MS in Computer Science

Sep 2025 – Jun 2027 (expected)

GPA: 3.95/4.0

Stanford, CA

- Coursework: Machine Learning (CS 229), NLP with Deep Learning (CS 224N), Deep Reinforcement Learning (CS 224R), Language Modeling from Scratch (CS 336), Machine Learning with Graphs (CS 224W)

Indian Institute of Technology Kanpur — B.Tech in Computer Science and Engineering

2017 – 2021

GPA: 9.94/10 · Graduated with Distinction · Academic Excellence Award (top 10%) in all four years

Kanpur, India

- Coursework: Data Structures & Algorithms, Advanced Algorithms, Mathematics for Computer Science I–III (probability & statistics, logic, discrete mathematics), Algorithmic Game Theory, Introduction to Machine Learning, Programming for Performance, Parallel Computing, Operating Systems, Mathematical Economics

EXPERIENCE

Upstart — Machine Learning Engineer Intern

Jun 2026 – Sep 2026

Personal-loan underwriting

Remote, USA

- Improving the accuracy, runtime, and cost of Upstart's machine learning models for personal-loan underwriting.

slice — AVP, Strategy & Operations

Apr 2024 – Jun 2025

Onboarding

Bengaluru, India

- Increased new user onboarding conversion by 2%** by eliminating friction points across the onboarding journey.
- Optimized the Video-KYC process**, cutting turnaround time by **30 seconds** through streamlined verification and automation.
- Enhanced the deduplication** system, cutting new duplicate-account onboardings to **near zero**.

Samsung Electronics HQ — Software Engineer

Sep 2021 – Mar 2024

System R&D Group

Suwon, South Korea

- Spearheaded integration and maintenance of Android touch-input kernel drivers across Samsung flagship and non-flagship devices.
- Restructured the Android service layer to process input matrices from multiple touch panels, enabling **gesture-recognition** machine learning models to run on dual-screen devices.
- Independently developed **linear-time algorithms for touch-gesture recognition**, pipelined from the kernel to the Android service layer.
- Integrated several AI-based recognition models into the Android service layer for flagship devices.

University of Washington — Research Intern

Jun 2019 – Jul 2019

Blockchain research with Prof. Sreeram Kannan

Seattle, WA

- Implemented a Rust-based blockchain application to empirically test research on increasing transaction throughput and reducing confirmation latency.

RESEARCH & PROJECTS

Video Preference Score — NeurIPS 2026 (under review)

Stanford

- Built a fully automated video preference score (reward model) that recreates Elo-based video-generation arenas without human raters; correlates highly with human consensus, generalizes zero-shot to unseen models, and improves the WAN-2.1 video generation model by **+51 Elo** via Flow-GRPO reinforcement learning.

Difficulty-Aware Curriculum Learning for RL Fine-Tuning of LLMs

CS 224R, Stanford

- Benchmarked curriculum learning across SFT, IPO, and RLOO post-training with an exact difficulty signal computed via dynamic programming over ~490k prompts; adaptive curricula lift SFT by **+7.1 pass@1** while offline objectives are largely order-invariant.

Casting the Graph — graph neural networks for cast recommendation

CS 224W, Stanford

- Built a heterogeneous IMDb graph (~37k movies, ~163k people) with BERT plot embeddings; trained R-GCN and HGT models for cast recommendation via link prediction (**AUC 0.76 vs 0.52** ALS baseline) and rating prediction via node regression (**RMSE 0.81 vs 1.04** Node2Vec + XGBoost).

PEFT for GPT-2: Paraphrase Detection and Sonnet Generation

CS 224N, Stanford

- Implemented the GPT-2 transformer architecture from scratch; parameter-efficient fine-tuning (LoRA/DoRA on KQV+MLP) recovers near full fine-tuning accuracy (0.882 vs 0.898) with **~7% of trainable parameters**.

SKILLS

Languages: Python, C++, C, Java, Bash

ML: PyTorch, PyTorch Geometric, NumPy, Pandas, vLLM, CUDA

Areas: LLM post-training (SFT, RLHF, GRPO), deep reinforcement learning, NLP, graph machine learning, generative model evaluation

Tools: Git, Perforce, Linux kernel, Android system software, R, MATLAB

AWARDS & TEACHING

- All India Rank 116** among 220,000 candidates, JEE Advanced (2017); **All India Rank 27** among 1.3 million candidates, JEE Mains (2017).
- Course Assistant, CS 231N: Deep Learning for Computer Vision, Stanford (Spring 2026).
- Tutor, Introduction to Programming, IIT Kanpur (2020–21); academic mentor for freshmen, IITK Counselling Service.